Abdulla Nasir Chowdhury

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Research interests: 3D computer vision, point cloud understanding, robustness and distribution shift, synthetic-to-real transfer, statistical evaluation of deep learning models.

Education

University of Tsukuba, Japan *Expected graduation: Mar. 2028*
M.Sc. in Computer Science, Computer Vision Laboratory, Apr. 2026 – Present
Research Student in the same laboratory, Apr. 2025 – Mar. 2026

Leading University, Sylhet, Bangladesh *2019 – 2023*
B.Sc. in Electrical and Electronic Engineering • CGPA 3.93 / 4.00

Additional coursework: Machine Learning Specialization, Stanford University and DeepLearning.AI (Coursera)

Research Experience & Publications

Graduate Researcher, Computer Vision Laboratory, University of Tsukuba *Apr. 2025 – Present*

- **First-author paper, WACV 2027 (Datasets Track):** *Do We Still Need Rotation-Invariant Architectures? A Statistical Robustness Benchmark for Point Cloud Classification* (arXiv:2611.03952). Benchmarked 10 architectures across 5 domains under pose, noise and occlusion with confirmatory hypothesis tests; rotation-invariant designs were the most pose-robust but among the least noise-robust in 4 of 5 domains.
- Showed that standard backbones trained with a simple invariance loss finish within 2.5 percentage points of rotation-invariant ones on unseen corruptions, by retraining all 10 architectures with 3 seeds each.
- **First-author paper, ACCV 2026 Workshop on Subspaces in the Era of Foundation Models:** *Subspace Representations for Synthetic-to-Real Transfer in 3-D Surface Inspection* (arXiv:2611.02744). Reached 0.957 point AUROC on pre-registered Real3D-AD scans and 78–81% category-mean object AUROC over 597 objects with no training on the benchmark, using one-class subspaces of registered templates on frozen features.

Research Assistant, Intex Research Lab *Sylhet, Bangladesh · 2024 – 2025*

- **First-author paper, ISACC 2025 (IEEE Xplore):** annotated and labelled Bangla characters and words for the Bangla CrossHair dataset and benchmarked PyTesseract and EasyOCR on it.

Research Assistant, EARL Research Network *Sylhet, Bangladesh · 2021 – 2023*

- **First-author paper, QPAIN 2025 (IEEE Xplore):** prototyped and simulated a real-time GPS and passenger tracking system (PTS) architecture.

Full publication list: homepage • Google Scholar

Software & Projects

pcrobust (PyPI, GitHub, MIT License) *2026*

- Python library for robustness evaluation of point cloud classifiers: corruption generators, 10 architectures, the statistical pipeline and 193 trained checkpoints, with a command that rebuilds the WACV 2027 paper's tables. Archived on Zenodo.

Real-Time Detection & Tracking (GitHub, live demo) *2026*

- Multi-object tracking pipeline with YOLO11 and ByteTrack, exported to ONNX and TensorRT with a measured CPU vs. GPU latency benchmark; includes a per-worker PPE compliance demo. Deployed on Google Cloud Run.

Point Cloud Inspection API (GitHub, live demo) *2026*

- Unsupervised 3D defect detection service needing no labelled defects: FPFH descriptors matched against a FAISS memory bank of defect-free parts, served with FastAPI in Docker and benchmarked on MVTec 3D-AD.

Technical Skills

- **Programming languages:** Python, C/C++, MATLAB
- **ML / Computer Vision:** PyTorch, scikit-learn, OpenCV, Open3D, FAISS, ONNX, TensorRT
- **Systems & Tools:** Linux, Git, Docker, FastAPI, Google Cloud Run
- **Statistics:** Hypothesis testing, ANOVA, effect-size analysis, experimental design

Awards & Languages

- **Award:** Japanese Government (MEXT) Scholarship
- **Languages:** English (IELTS 8.0, CEFR C1), Japanese (JLPT N4, conversational)